$s = 1$, and hence

$$\tau_0 = \frac{1}{2 - \sqrt[3]{2}}, \qquad \tau_1 = -\frac{\sqrt[3]{2}}{2 - \sqrt[3]{2}}.$$

The overall scheme is then three iterations of velocity Verlet, using stepsizes $\tau_0 h$, $\tau_1 h$ and $\tau_0 h$ respectively. We write this with subindices α, β to indicate the intermediate stages.

Yoshida fourth-order scheme (velocity Verlet):

$$\begin{aligned}
\boldsymbol{P}_\alpha &\coloneqq \boldsymbol{p} - (\tau_0 h/2)\, \nabla U(\boldsymbol{q}), \\
\boldsymbol{Q}_\alpha &\coloneqq \boldsymbol{q} + (\tau_0 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}_\alpha, \\
\boldsymbol{P}_\alpha &\coloneqq \boldsymbol{P}_\alpha - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}_\alpha), \\
\boldsymbol{P}_\beta &\coloneqq \boldsymbol{P}_\alpha - (\tau_1 h/2)\, \nabla U(\boldsymbol{Q}_\alpha), \\
\boldsymbol{Q}_\beta &\coloneqq \boldsymbol{Q}_\alpha + (\tau_1 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}_\beta, \\
\boldsymbol{P}_\beta &\coloneqq \boldsymbol{P}_\beta - (\tau_1 h/2)\, \nabla U(\boldsymbol{Q}_\beta), \\
\boldsymbol{P} &\coloneqq \boldsymbol{P}_\beta - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}_\beta), \\
\boldsymbol{Q} &\coloneqq \boldsymbol{Q}_\beta + (\tau_0 h)\, \boldsymbol{M}^{-1} \boldsymbol{P}, \\
\boldsymbol{P} &\coloneqq \boldsymbol{P} - (\tau_0 h/2)\, \nabla U(\boldsymbol{Q}).
\end{aligned} \tag{3.8}$$

The equations could easily be written in a simplified form, combining several of the steps. This scheme requires three new evaluations of the force ∇U per iteration, making it significantly more expensive than the vanilla second-order Verlet method. We may use this fourth-order scheme (now $s = 2$) to then find a scheme of order $2s + 2 = 6$.

We compare the symplectic methods developed so far in Fig. 3.2, for $t \in [0, 100]$ using planar phase space and the Lennard-Jones oscillator (the one degree-of-freedom problem with Hamiltonian $H(q, p) = p^2/2 + \varphi_{\mathrm{LJ}}(q)$). As anticipated, there is no visible drift in the Hamiltonian as the simulation time increases. In each case, we observe an oscillation in the fluctuation of the total energy. Additionally we compute trajectories propagated at different stepsizes, and plot the maximum deviation

$$\max |H(q_k, p_k) - H(q_0, p_0)|$$

over the entire trajectory. We can see that for an order s method, the maximum deviation in the Hamiltonian is of order s, as is expected from the analysis. Yoshida fourth-order schemes are tested using both the position and velocity Verlet schemes